Computer Graphics

CSCI 490J, CSCI 631

Project 4

Solid Polygons

Summary

The purpose of this assignment is to do polygon scan conversion and clipping.

Program Requirements

Your program should be able to

- draw cubes, spheres, cones, cylinders, and arbitrary polygon sets as solid objects.
- properly clip solid objects in x, y and z.
- properly display depth ordering relationships between objects.

Additional scene files are available for this project.

Things That Need to be Implemented and Modified — What to Do

This is a list of data structures and routines that need to be implemented or modified, and a suggested implementation order.

New Data Structure

- Get a copy of the structures which are needed to properly perform polygon clipping and eventual lighting calculations. There is a new type for an attributed point.

Pipeline

- Make a new pipeline to handle polygons. Get a copy of a C / pseudocode implementation to guide you.
- The semantics of the move/draw flag passed to the polygon pipeline has changed. All vertices of a polygon to be drawn should have the flag set to `move`, except the last, which should be `draw`. In this case, `draw` has the meaning "Draw the polygon consisting of all the vertices received since the last `draw`." Unlike the line pipeline, the last vertex of the polygon is *not* a repeat of the first. The clipping and scan conversion routines take care of this automatically.

Polygon Scan Conversion

Use the polygon fill code found in this C/C++ pseudocode.

Polygon Clipping

Use the Sutherland-Hodgman clipping code found in this pseudocode.

Objects

- Change your `cube`, `cone`, `cylinder`, `sphere`, `disk`, and `polyset`routines to run polygons through the polygon pipeline instead of the line pipeline. The points passed into the polygon pipeline should be attributed points. For now, only the x, y, z, and w coordinates should be set for each point before it is passed through the pipeline.

  Remember, the semantics of the move/draw flag passed to the polygon pipeline has changed. See above for details.

Suggested Implementation Order

- Implement the polygon pipeline function with empty functions for the clipper and scan conversion.
- Rewrite your cube drawing routine to use the polygon pipeline. Only send in one face of the cube for now. Comment out the rest.
- If you have not already done so, implement a function to plot a single point given device coordinates in x, y, and z and an rgb color value. This function should do the z buffer testing and plot the pixel.
- Implement the scan conversion using the routines explained above. The scan conversion should make use of your single point plotting routine from the previous step. When this step works properly, you should be able to make a colored square from a scene file using the Cube command. Rerun some of the scene files from previous assignments that don't involve clipping.
- Enable all of the other faces of your cube, using the polygon pipeline.
- Replace your sphere, cone, cylinder, disc, and polyset routines.
- Implement polygon clipping using this pseudcode.

Pass off

As always, make a scene of your own, and show the resulting image at passoff. No need to do anything elaborate. Solid versions of previous wireframe scenes will usually be excellent, although some tweaking may be in order.

Wrap up the project into a single file as explained on the website and e-mail it to the address given on the website.